

Dingyi Zhuang

Website: zhuangdingyi.github.io/
Google Scholar

LinkedIn
Email : dingyi.zhuang97@gmail.com
Mobile : +1-617-803-4153

SUMMARY

Sr. Machine learning researcher with a Ph.D. from MIT, focused on spatiotemporal prediction, human behavior modeling, and decision-making under uncertainty. My research spans probabilistic forecasting, graph neural networks, uncertainty calibration, and multimodal spatial reasoning, with publications at ICLR, KDD, EMNLP, and AAAI, as well as workshops at NeurIPS and The Web Conference. In industry, I develop forecasting and agentic decision-support systems that integrate predictive models, causal analysis, and operational constraints. Broadly, I aim to build reliable learning systems that anticipate dynamic environments, quantify uncertainty, and support robust prediction and planning for autonomous systems.

EDUCATION

- Massachusetts Institute of Technology** Cambridge, MA
Ph.D. in Transportation Engineering, Machine Learning / AI Systems Focus Sep. 2021 – April. 2026
 - Research focus: reliable ML evaluation, uncertainty calibration, graph/time-series learning, and evaluation of agentic systems.
 - Dissertation: uncertainty quantification, calibration, and fairness-aware evaluation for sparse and high-stakes systems.
- McGill University** Montreal, Canada
Master in Transportation Engineering Sep. 2019. - May. 2021
- Shanghai Jiao Tong University** Shanghai, China
Bachelor of Science in Mechanical Engineering (Tsien Hsue-Shen Class, Top 5%) Sep. 2015 - Jul. 2019

WORK EXPERIENCE

- TikTok Inc.** San Jose, CA
Senior Machine Learning Engineer April. 2026 - Now
 - Building agentic workflows for supply-chain planning, integrating demand forecasting, inventory optimization, causal simulation, and human-in-the-loop decisions with traceable evidence and safe fallbacks.
 - Developing time-series models for sparse SKU-level warehouse demand, with hierarchical forecasting, cold-start handling, and rolling backtests across regions and ZIP3s.
 - Building a self-improving Causal Effect Bank for demand-driver and price-elasticity discovery, using hierarchical estimation, uncertainty-aware validation, and fallback gates to power sales and inventory what-if simulations.
- Bosch Center for Artificial Intelligence** Sunnyvale, CA
Research Intern May. 2025 - Aug. 2025
 - Built plug-and-play reasoning modules for online map generation in autonomous driving, improving topology-aware evaluation on OpenLaneV2 with minimal training and inference overhead.
 - Designed graph-structure refinement and contrastive representation learning methods to improve structural consistency, failure-case robustness, and map-level reasoning quality.
- Morgan Stanley** New York City, NY
Machine Learning Research Intern Jun. 2024 - Aug. 2024
 - Designed imbalance-aware heterogeneous GNNs for fraud-account detection on semi-bipartite financial transaction networks; methods were implemented into production-facing workflows for related graph-structured problems.
 - Built analysis dashboards and evaluation pipelines for transaction, email, and financial-network graphs, supporting error analysis, thresholding, and model diagnostics under noisy class-imbalanced signals.

SELECTED PUBLICATIONS (40+ PAPERS, CITATION 1300+)

- Dingyi Zhuang***, Xiaoqi Wang*, David Paz, Wenbin He, Yucai Bai, Peng Zhou, Rui Zhang, Jinhua Zhao, Liu Ren, Refine Connections, Close the Gap: A Reliable Enhancement Framework for Driving Scene Topology, NeurIPS 2026 under review.
- Dingyi Zhuang**, Chonghe Jiang, Yunhan Zheng, Shenhao Wang, Jinhua Zhao, GETS: Ensemble Temperature Scaling for Calibration in Graph Neural Networks, *ICLR 2025*, *Spotlight*. [link](#)
- Xinyu Chen, **Dingyi Zhuang**, HanQin Cai, Shenhao Wang, Jinhua Zhao, Dynamic Autoregressive Tensor Factorization for Pattern Discovery of Spatiotemporal Systems, IEEE Transactions on Pattern Analysis and Machine Intelligence. [link](#)
- Dingyi Zhuang**, Hanyong Xu, Xiaotong Guo, Yunhan Zheng, Shenhao Wang, Jinhua Zhao, Mitigating Spatial Disparity in Urban Prediction Using Residual-Aware Spatiotemporal Graph Neural Networks: A Chicago Case Study, *WWW 2025*, *Best Paper Award* at *WebST* Workshop. [link](#)
- Dingyi Zhuang**, Yuheng Bu, Guang Wang, Shenhao Wang, Jinhua Zhao, SAUC: Sparsity-Aware Uncertainty Calibration for Spatiotemporal Prediction with Graph Neural Networks, TGL Workshop at *NeurIPS 2023*. [link](#)
- Dingyi Zhuang**, Shenhao Wang, Haris Koutsopoulos, Jinhua Zhao, Uncertainty Quantification of Sparse Trip Demand Prediction with Spatial-Temporal Graph Neural Networks, *The 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (SIGKDD 2022)*. [link](#)